

Mirror symmetry for hypersurface singularities

1. Modality

germ of a holomorphic function

$$f: (\mathbb{C}^n, 0) \rightarrow (\mathbb{C}, 0)$$

assume $0 \in \mathbb{C}^n$ isolated singularity of f with Milnor number $\mu = \mu(f, 0)$.

~~Let $\Lambda \subset \mathbb{C}^m$ be a Milnor fibre~~

Versal unfolding $f_\lambda: (\mathbb{C}^n, 0) \rightarrow (\mathbb{C}, 0)$

for $\lambda \in \Lambda$ an open nbhd $0 \in \Lambda \subset \mathbb{C}^m$

$\exists$ stratification of Λ by $\mu(f_\lambda, 0)$

Example (A_2) $f(x) = x^3: \mathbb{C} \rightarrow \mathbb{C}$

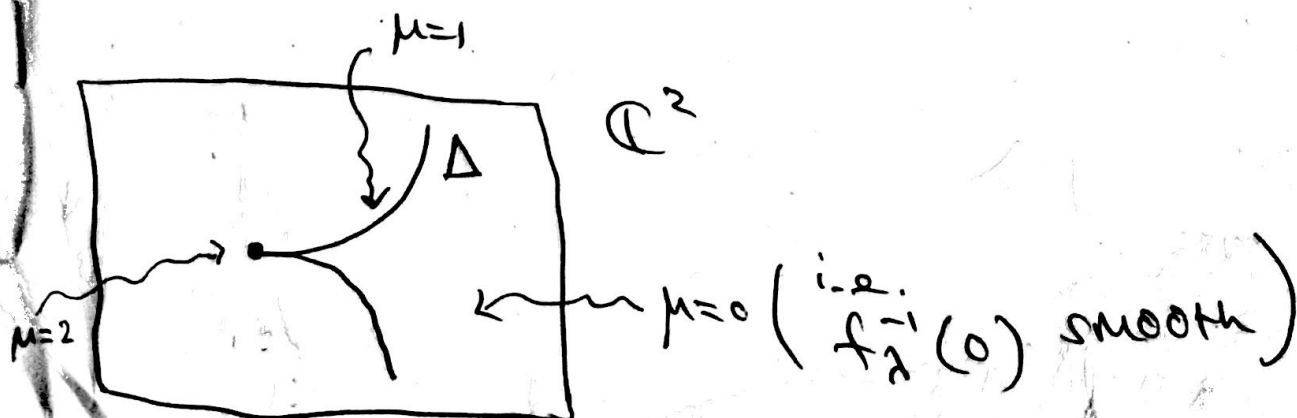
$$f_\lambda(x) = x^3 + ax + b, \quad \lambda = (a, b) \in \Lambda = \mathbb{C}^2$$

$$0 \subset \Delta = \{4a^3 + 27b^2 = 0\} \subset \Lambda = \mathbb{C}^2$$

$$\mu = 2$$

$$\mu \geq 1$$

$$\mu \geq 0$$



Defⁿ: modality $M(f) = \dim_{\mathbb{C}} \{ \lambda \in \underline{\Lambda} : \mu(f, \lambda) = \mu \}$
 = dimension of deepest stratum.

2. Classification

$\exists$ classification of isolated hypersurface singularities of modality $M(f) \leq 2$ up to stable equivalence (Arnold '74)

$M(f) = 0$ (= simple singularity) ADE

$$A_k \quad x^{k+1} + y^2 + z^2$$

$$D_k \quad x^2 y + y^{k-1} + z^2$$

$$E_6 \quad x^3 + y^4 + z^2$$

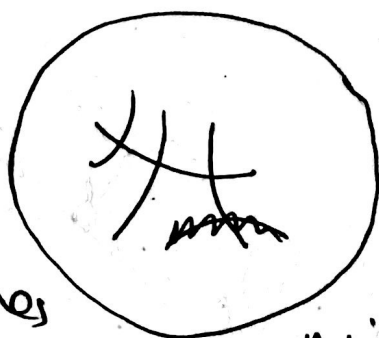
$$E_8 \quad x^3 + y^5 + z^2$$

$$E_7 \quad x^3 + xy^3 + z^2$$

Set $X = f^{-1}(0)$,
 singularity

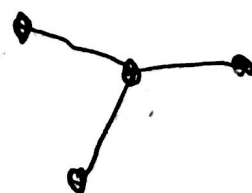
$M = f^{-1}(\epsilon)$
 Milnor fibre

-2
 curves



minimal resolution

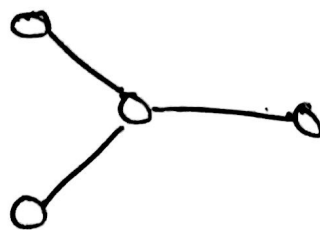
dual graph
 $\rightsquigarrow$



Dynkin diagram

Intersection form
on $H_2(M)$

$$\mathbb{Z}S_1 \oplus \dots \oplus \mathbb{Z}S_\mu$$



$$S_i \cdot S_i = -2$$

$$S_i \cdot S_j = \begin{cases} 1 & \text{if } i=j \\ 0 & \text{if } i \neq j \end{cases}$$

In fact $\tilde{X} \xrightarrow[\mathbb{C}^\infty]{\cong} M$

$M(f) = 1$ ($\because$ unimodal singularity)

Three-index series

$$\frac{1}{p} + \frac{1}{q} + \frac{1}{r} \leq 1$$

$$x^p + y^q + z^r + axyz$$

$$a \in \mathbb{C}$$

+ 14 exceptional cases

eg K_{13}

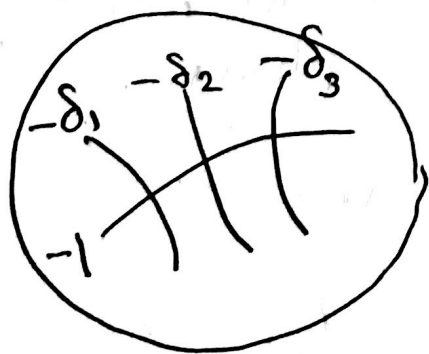
$$x^3 + xy^5 + z^7 + ay^8$$

$$(2, 4, 5)$$

$\mathbb{Z}_{11}$

$$x^3y + y^5 + z^2 + axy^4$$

$$(2, 3, 8)$$



$\tilde{X}$

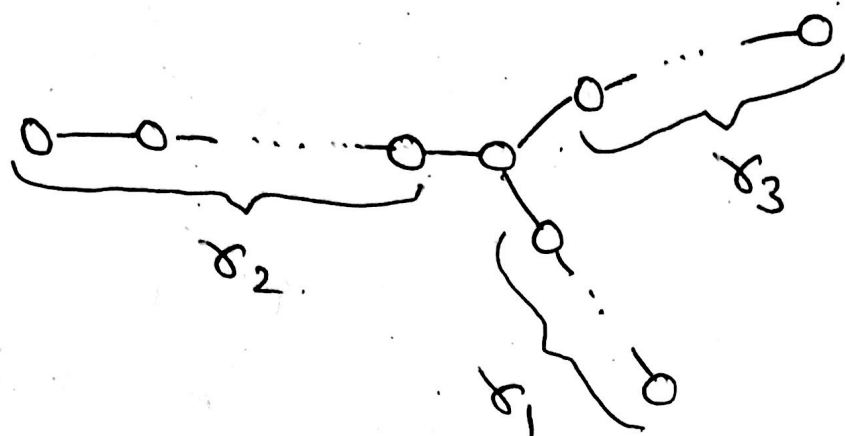
$$H_2(M) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \oplus T(r)$$

$$r = (r_1, r_2, r_3)$$

Gabrielov #s

$\delta_1, \delta_2, \delta_3$

Dolgachev #s



$$T(\delta_1, \delta_2, \delta_3)$$

Arnold's strange duality (1974)

$\exists$ involution on the

$$14 = \underbrace{6}_{\curvearrowright} + \underbrace{(4+4)}_{\curvearrowright}$$

exceptional unimodal

singularities which exchanges $(\delta_1, \delta_2, \delta_3)$

with $(\delta_1, \delta_2, \delta_3)$. E.g. $K_{13} \leftrightarrow Z_{11}$

"at the present stage the theory of singularities is at an experimental stage" aka wtf?

3. Homological mirror symmetry
for hypersurface singularities

$$(\mathbb{C}^n, 0) \xrightarrow{f} (\mathbb{C}, 0) \quad \text{"LG model" in physics}$$

"A-branes": Fukaya-Seidel category $\mathcal{D}_A$

"B-branes": Singularity category / matrix factorisations $\mathcal{D}$

Note : $K_0(D_A) \cong H_{n-1}(M, \mathbb{Z})$

and symmetrised Euler form $-X(a,b) - X(b,a) \leftrightarrow$ intersection form

In quasi-hom. cases D_B is closely related to $D^b \text{Coh} \left(\left[\frac{X \setminus \{0\}}{\mathbb{C}^*} \right] \right)$.

(In exceptional unimodal cases this is a weighted proj. line ~~with~~ with orbifold points $\frac{\mathbb{Z}}{8}, \frac{\mathbb{Z}}{2}$)

Expectation : Arnold strange duality lifts to (theorem?)

eg $D_A(K_{13}) = D_B(Z_{11})$
 $D_B(K_{13}) = D_A(Z_{11})$

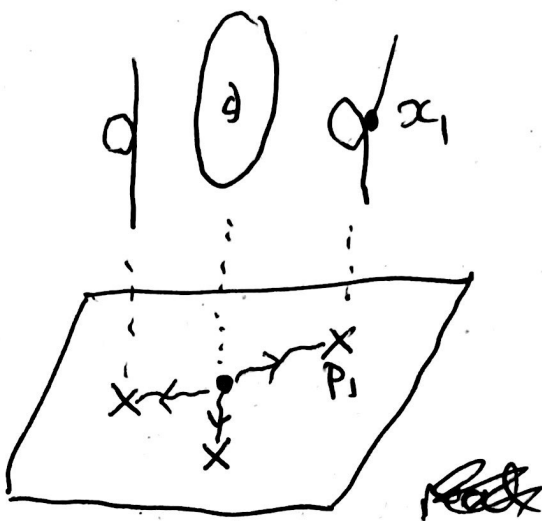
f. Fukaya-Seidel category

: Choose $\lambda \in \Lambda$ so that $\pi = f_\lambda$ has simple critical points $x_1, \dots, x_\mu$ with distinct critical values $\theta_1, \dots, \theta_\mu \in \mathbb{C}^*$

$$\mathbb{P}^n = Y$$

$$\downarrow f_\lambda = \pi$$

$\mathbb{P}$



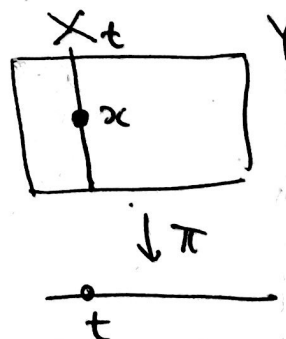
Choose an exact ~~real~~ 2-form Ω on $\mathbb{P}^n$

which restricts to a symplectic form on each fibre $X_t = \pi^{-1}(t)$

Choose paths $\gamma_i : [0, 1] \rightarrow \mathbb{P}$
 $\gamma_i(0) = 0, \gamma_i(1) = t_i$

Note : Ω defines a non-linear connection on π .

$$\begin{array}{ccc} T_{x_t} X_t & \hookrightarrow & T_x Y \\ & \searrow \omega & \downarrow \rho \\ & & T_x^* Y \end{array} \quad \begin{array}{c} \xrightarrow{\quad} \\ T_x^* X_t \end{array}$$



$$T_x Y = \underbrace{T_{x_t} X_t}_{\text{vertical}} \oplus \underbrace{\ker \rho}_{\text{horizontal}}$$

Use this to parallel transport vanishing cycles to Lagrangians $L_i \subset X = X_0$

$$L_i = \left\{ x \in X_0 : PT_{x_i}(x) = x_i \right\}$$

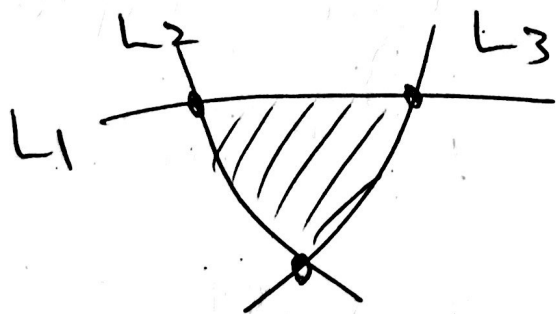
Define an A_∞ category with objects $L_1, \dots, L_\mu$ (assumed to meet transversally)

$$\text{Hom}(L_i, L_j) = \begin{cases} \mathbb{C}^{\#(L_i \cap L_j)} & i < j \\ \mathbb{C} & i = j \\ 0 & \text{otherwise} \end{cases}$$

A_∞ -products

given by counting holomorphic discs

$$\text{Hom}(L_1, L_2) \times \text{Hom}(L_2, L_3) \rightarrow \text{Hom}(L_1, L_3)$$



Key-point: Resulting Δ -category has an exceptional collection $(L_1, \dots, L_\mu)$ (really these are the Lefschetz thimbles)

and changing the paths γ_i corresponds to mutation. (both controlled by braid group $\pi_1(\text{Config}_n(\mathbb{C}))$)

Result : D_A independent of all choices

Conjecture : $\text{Stab}(D_A) \cong \Lambda$

$\downarrow$
 $H^n(M, \mathbb{C})$ ← periods

Ueda et al

$\cdots \rightarrow P_0 \xrightarrow{d^0} P_1 \xrightarrow{d^1} P_2 \rightarrow \cdots$ ← free A -modules
 $\mathbb{C}[x_1, \dots, x_n]$

$d^i \circ d^{i-1} = 0$ cochain complex

homotopy

$$x^3 + xy^2 + z^2$$